Risk Flagging System Design

Objective

Provide recruiters with clear integrity risk indicators based on browser activity, behavioral signals, and environmental observations during AI interviews.

Risk Categories

Integrity Score	Risk Level
75–100	Low Risk
50–74	Moderate Risk
Below 50	High Risk

Real-Time Warning Rules

Event	Warning
Tab switches > 2	Please stay on the interview screen
Multiple voices detected	Ensure you are alone during the interview
Focus loss > 3	Please focus on the interview
Frequent gaze deviation	Maintain attention on the interview

Flagging Rules

Event	Flag
Excessive tab switching	High Tab Switching
Browser focus loss	Screen Focus Loss
Multiple speakers	Multiple Voices Detected
Repeated gaze deviation	Frequent Gaze Deviation

Pattern Recognition

The system identifies combinations of events that may indicate potential malpractice.

Examples:

- Repeated tab switching → Possible use of external resources
- Continuous voice detection → Possible external assistance
- Gaze deviation with long pauses → Possible note consultation
- Focus loss with delayed responses → Possible multitasking

Recruiter Dashboard Output

```
{
    "candidate_id": "C4001",
    "integrity_score": 68,
    "risk_level": "Moderate Risk",
    "flags": [
        "High Tab Switching",
        "Frequent Gaze Deviation"
    ],
    "warnings": [
        "Please stay on the interview screen."
    ]
}
```

Integration Flow

```
Interview Start
        |
        ▼
Browser Activity Monitoring
        |
        ▼
Audio Monitoring
        |
        ▼
Behavior Analysis
        |
        ▼
Signal Aggregation
        |
        ▼
Integrity Risk Calculation
        |
        ▼
Warning & Flag Generation
        |
```

```
        ▼
Recruiter Dashboard
```

Ethical Guidelines

- Candidate consent required
- No facial recognition
- No screen recording
- Metadata-only analysis
- Transparent integrity scoring

Benefits

- Reduces interview malpractice
- Supports fair hiring decisions
- Provides explainable recruiter insights
- Preserves candidate privacy

Future Improvements

- AI-based anomaly detection
- Voice identity verification
- Adaptive behavioral learning
- Cross-platform monitoring